DAC-LOOP: DISTILLATION-AWARE COMPRESSION FOR REMOTE SENSING CNN

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Problem Statement & Motivation

- **Large CNNs**
  ✓ High accuracy
  ✗ Too large for edge devices
- **Need for small CNNs that are**
  ✓ Fast
  ✓ Memory efficient
  ✓ Accurate

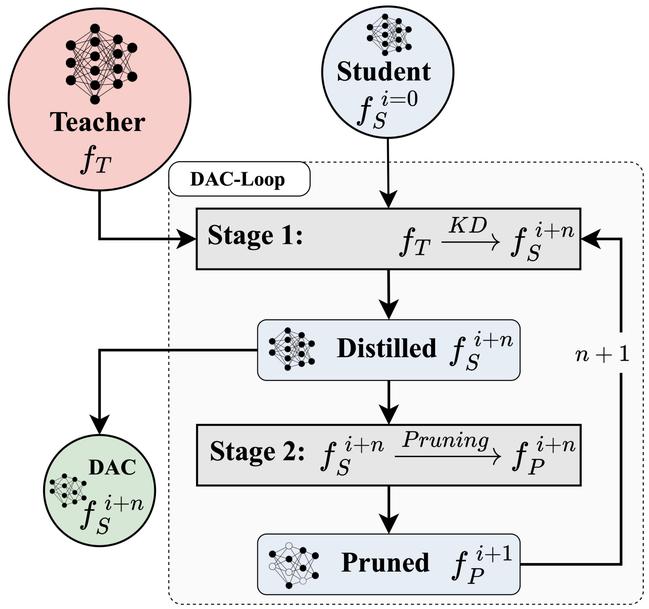
Can we compress a CNN by over 90% while retaining almost all of its accuracy?

Pruning removes redundant params / channels

Knowledge Distillation (KD) transfers a teacher's knowledge to a smaller student

- Combining mechanisms into an iterative compression Loop (DAC-Loop)

The DAC-Loop

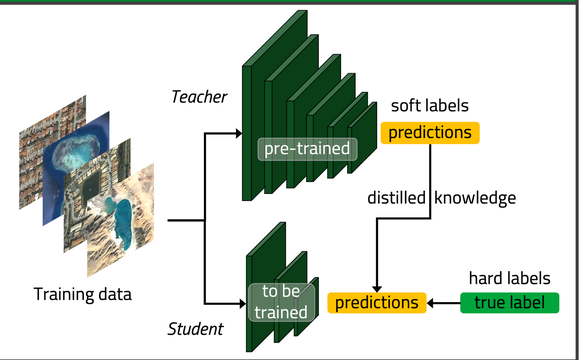


Prune → Recover with KD → Repeat
Each cycle progressively compresses the student, while guidance from the fixed teacher restores performance after pruning.

Next DAC-iteration (**n + 1**)

Knowledge Distillation (KD) – Stage 1

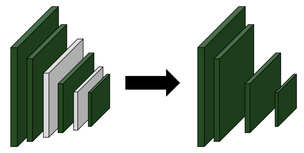
- Train the compact student using ground-truth labels and the teacher's soft predictions
- Soft predictions transfer richer class relationships than hard labels alone
- Reuse the fixed teacher after each pruning step to recover lost accuracy



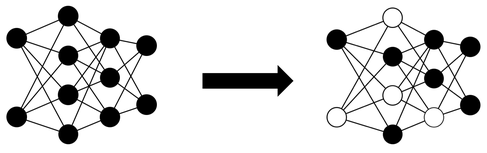
Pruning – Stage 2

➢ **Iteratively remove low-importance channels and weights**

Structured channel pruning
Removes entire convolutional channels, producing a smaller architecture with lower dense computation.

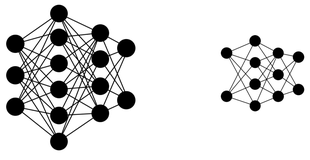


Weight magnitude pruning
Zeros individual low-magnitude weights, increasing sparsity without changing the network architecture.

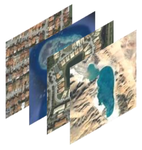


Models & Datasets

Models
➢ ResNet-50 Teacher (24.6 M)
➢ ResNet-18 Student (11.2 M)



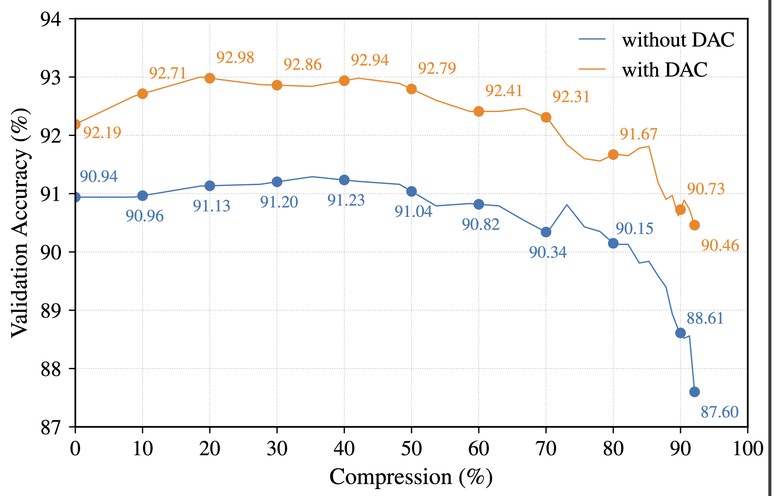
Datasets
➢ EuroSAT
➢ NWPU-RESISC45



Compression w/o DAC-Loop

Experiment Configuration
➢ ResNet-50 Teacher (24.6 M)
➢ ResNet-18 Student (11.2 M)
➢ NWPU-RESISC45
➢ Iterative pruning schedule
➢ Final Model size → 0.88 M

Schedule
1. KD with pretrained teacher
2. Channel pruning
3. Retrain with KD
4. Repeat step 2 & 3 for 25x



Main Contributions

Iterative KD + pruning framework for compact, high-performance CNNs

92% smaller Model
only 1.7% accuracy loss!

GitHub

Weight-Space Labs